

ECO102: Introduction to Macroeconomics

Lecture 12: Finance, Saving, and Investment

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Outline

- ❑ [Key Definitions](#)
- ❑ [Financial Capital Markets](#)
- ❑ [Financial Institutions](#)
- ❑ [Interest Rates and Asset Prices](#)
- ❑ [The Loanable Funds Market](#)
- ❑ [The Loanable Funds Market: The Real Interest Rate](#)

Key Definitions

- ***Finance***: The term is used to describe the activity of providing the funds that finance expenditures on capital
- ***Money***: What we use to pay for goods and services and factors of production and to make financial transactions
- ***Physical capital*** are the tools, instruments, machines, buildings, and other items that have been produced in the past and that are used today to produce goods and services
- The funds that firms use to buy physical capital are called ***financial capital***
- When economists use the term capital, they mean physical capital
- ***Investment*** increases the quantity of capital and ***depreciation*** decreases it
- The total amount spent on new capital is called ***gross investment***
- **Net investment equals Gross Investment minus Depreciation**

Key Definitions

- **Wealth** is the value of all the things that people own
- **Income** is the amount a person receives during a given time period from supplying the services of the resources they own
- Wealth is the **stock** of asset, and income is the **flow** of earnings (Stock vs Flow)
- **Saving** is the amount of income that is not paid in taxes or spent on consumption goods and services, and it increases wealth
- Wealth also increases when the market value of assets rises, called **capital gains**; and decreases when the market value of assets falls called **capital losses**

Financial Capital Markets

- Saving is the source of the funds that are used to finance investment, and these funds are supplied and demanded in three types of financial markets:
- ***Loan Markets:*** Used as a method of short-term financing to buy inventories (business), or to purchase big ticket items (households). Banks are one of the primary agents in the loan market

What is a mortgage?

- ***Bond Markets:*** A bond is a promise to make specified payments on specified dates.

Bonds issued by firms and governments are traded in the bond market. The term of a bond might be long (decades) or short (just a month or two). Bangladesh Bank issues Treasury Bonds and Bills

Another type of bond is a ***mortgage-backed security***, which entitles its holder to the income from a package of mortgages

Financial Capital Markets

- ***Stock Markets:*** A stock is a certificate of ownership and claim to the firm's profits
Unlike a stockholder, a bondholder does not own part of the firm that issued the bond
A stock market is a financial market in which shares of stocks of corporations are traded

Financial Institutions

- A ***financial institution*** is a firm that operates on both sides of the markets for financial capital. The financial institution is a borrower in one market and a lender in another
- ***Commercial banks***: Financial institutions that accept deposits, provide payment services, and make loans to firms and households
- ***Government-sponsored mortgage lenders***: Financial institutions that buy mortgages from banks, package them into mortgage-backed securities, and sell them, i.e. Fannie Mae and Freddie Mac
- ***Pension funds***: Pension funds are financial institutions that use the pension contributions of firms and workers to buy bonds and stocks
- ***Insurance companies***: Insurance companies enable households and firms to cope with risks and shocks. They receive premiums from their customers and pay back their claims in case of any risks and shocks

Insurance companies use the funds they have received but not paid out as claims to buy bonds and stocks on which they earn interest income

Interest Rates and Asset Prices

- Stocks, bonds, short-term securities, and loans are collectively called ***financial assets***
- The interest rate (the rate of return) on a financial asset is the interest received expressed as a percentage of the price of the asset

Let's consider a bond that promises to pay its holder \$5 a year forever. What is the rate of return, the interest rate, on this bond?

If you buy this bond for \$50, the interest rate would be:

$$\text{Interest rate} = (\$5 / \$50) \times 100 = 10\%$$

Or 10% per year

Think about what happens if you buy this bond for \$200?

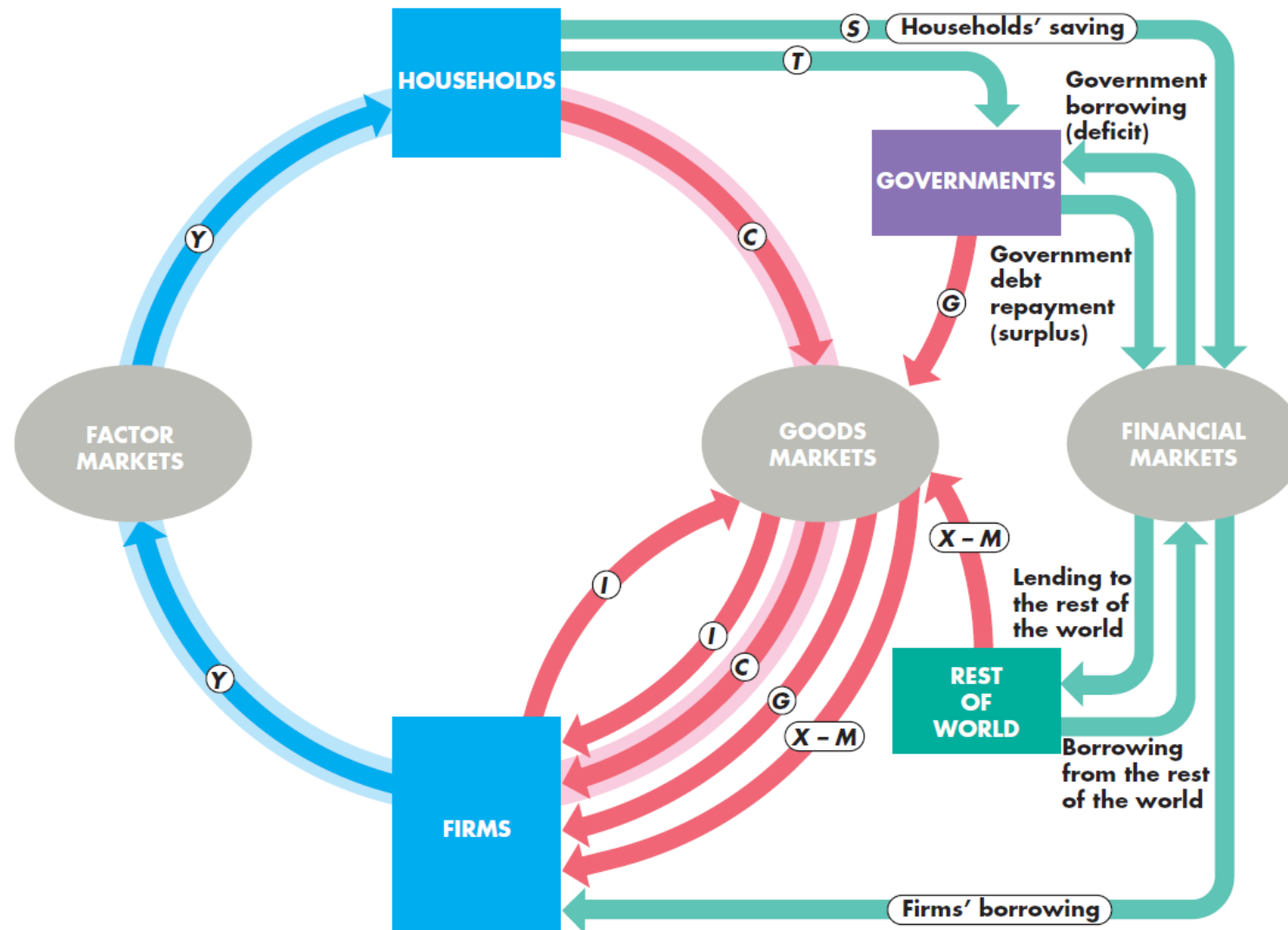
The Loanable Funds Market

- ***The loanable funds market*** is the aggregate of all the individual financial markets
- The flow of funds that finance investment come from three sources:
 - Household Saving
 - Government Budget Surplus
 - Borrowing from the rest of the world
- Households' income, Y , is spent on consumption goods and services (**C**), saved (**S**), or paid in net taxes (**T**)
- Net taxes are the taxes paid to governments minus the cash transfers received from governments
- Comparing with the equation for National Income Accounts we get:

$$I = S + (T - G) + (M - X)$$

What do each of these components mean?

The Loanable Funds Market



The Loanable Funds Market: The Real Interest Rate

- The ***nominal interest rate*** is the number of dollars that a borrower pays and a lender receives in interest in a year expressed as a percentage of the number of dollars borrowed and lent
- The ***real interest rate*** is the nominal interest rate adjusted to remove the effects of inflation on the buying power of money
- The real interest rate is approximately equal to the nominal interest rate minus the inflation rate
- The real interest rate is the opportunity cost of loanable funds
- The real interest paid on borrowed funds is the opportunity cost of borrowing
- And the real interest rate forgone when funds are used either to buy consumption goods and services or to invest in new capital goods is the opportunity cost of not saving or not lending those funds

Question

***THINK ABOUT HOW INTEREST RATES DIFFER ACROSS DIFFERENT
SECURITIES WITH VARYING RATE OF RETURN?***